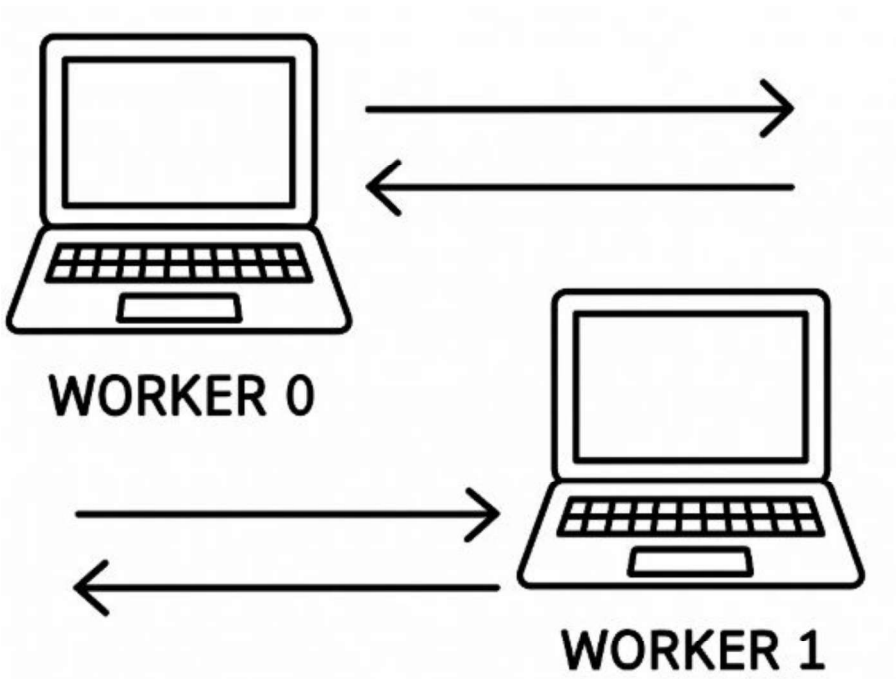
Projekt 2

Dávid Cauner, Matej Horský

Trénovanie prístup 1

- 2 počítače naraz
  - worker 0 = main
  - worker 1
- worker 0 zadáva úlohy
- dva identické projekty
- oba Pc na GPU 4060

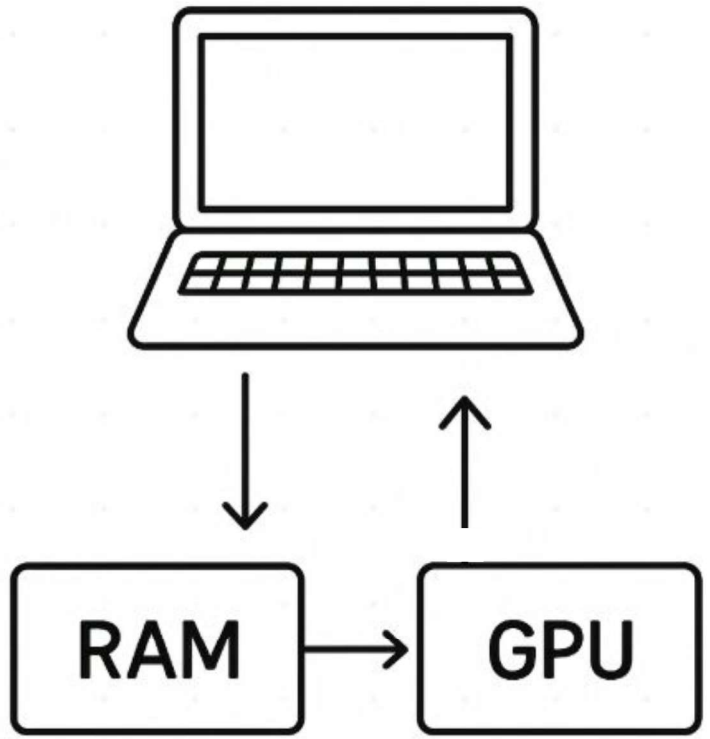
- nevýhoda
  - obmedzenie rýchlosti internetom



Trénovanie prístup 2

- jeden počítač na GPU
- načítanie datasetu na RAM

- nevýhoda
  - spapalo veľa ramky

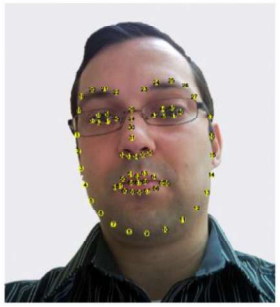


Úloha 1. Rozpoznávanie dobre a zle nasadených rúšok

DATASET



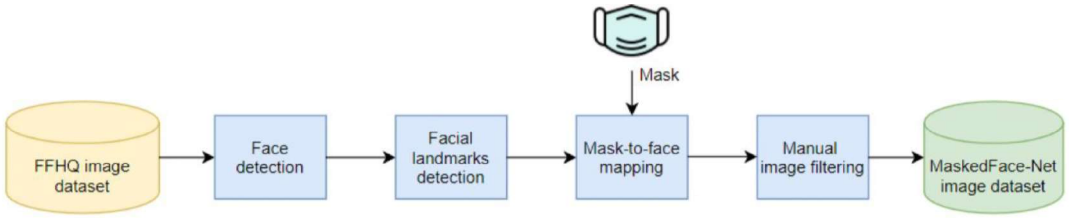
(a) Face image (first author).



(b) Detected landmarks.



(c) Annotated landmarks.



GitHub

GitHub - cabani/MaskedFace-Net: MaskedFace-Net is a dataset of huma...

MaskedFace-Net is a dataset of human faces with a correctly and incorrectly worn mask based on the dataset Flickr-Faces-HQ (FFHQ). - GitHub - cabani/MaskedFace-Net: MaskedFace-Net is a dataset of ...



Adnane Cabani, MaskedFace-Net – A dataset of correctly/incorrectly masked face images in the context of COVID-19,Smart Health,Volume 19,2021,100144,ISSN 2352-6483, https://doi.org/10.1016/j.smhl.2020.100144.

DATASET klasifikácia na **correct** a **incorrect**



MobileNetV2

- Fáza 1. rozmrazené vrstvy
- Fáza 2. rozmrazené vrstvy
- ostatné vrstvy boli zmrazené

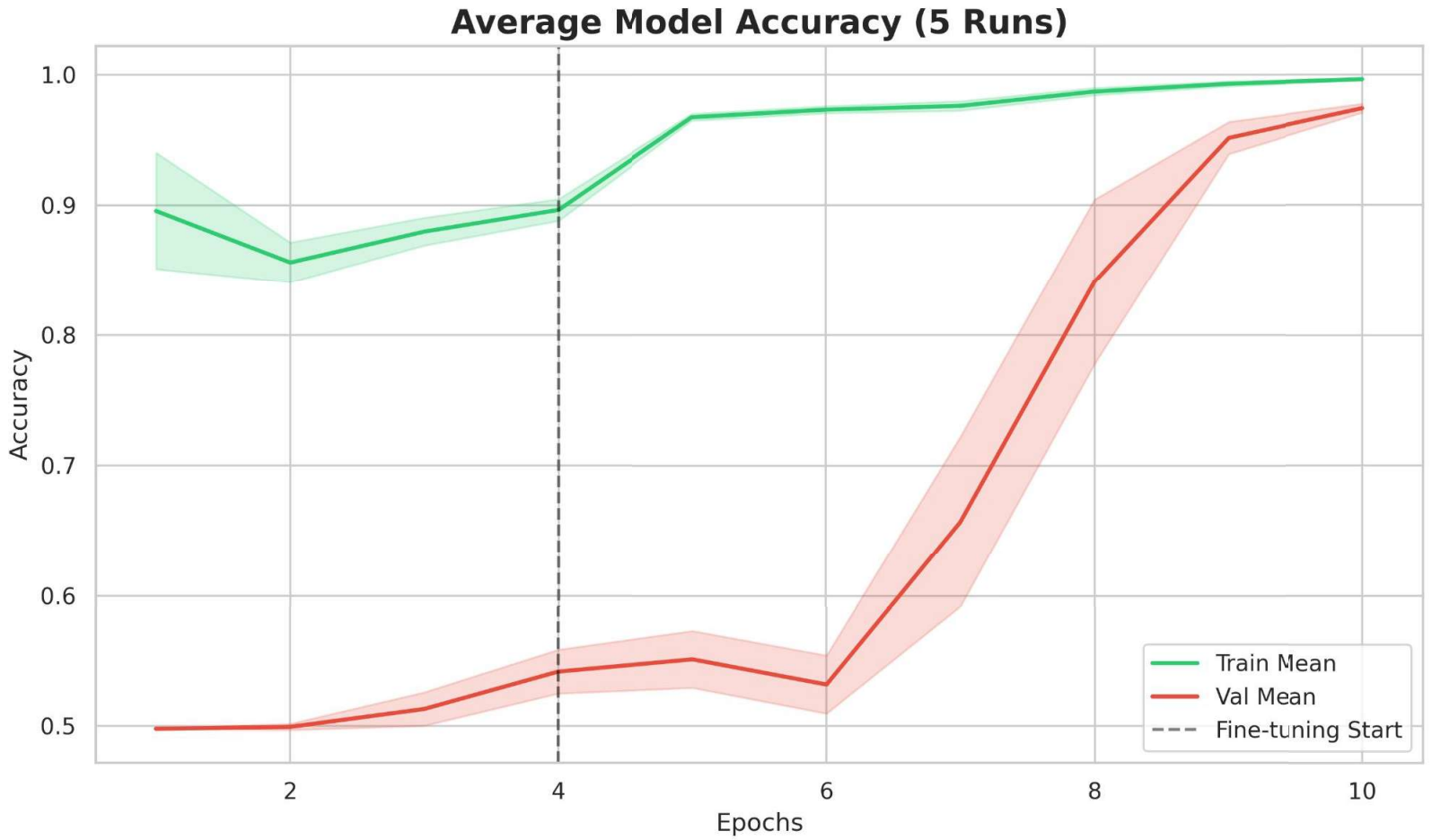
(0) Conv2D(3→32)
(1) BatchNorm
(3) Bottleneck (t=1,c=16)
(4) Bottleneck (t=6,c=24) ×2
(5) Bottleneck (t=6,c=32) ×3
(6) Bottleneck (t=6,c=64) ×4
(7) Bottleneck (t=6,c=96) ×3

(8) Bottleneck (t=6,c=160) ×3 → F2
(9) Bottleneck (t=6,c=320) → F2

(10) Conv2D(320→1280) → F2
(11) BatchNorm

(14) Dropout(0.3)
(15) Dense(2, softmax) → F1, F2

Výsledky


Average Model Accuracy (5 Runs)
Accuracy
Epochs
1.0
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10
Train Mean
Val Mean
Fine-tuning Start

Výsledky 96.9%

Senzitivita: 100%
Špecificita: 93.75%

